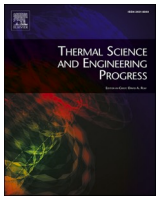
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The role of the phase-change material properties, porosity constant, and their combination for melting problems using the enthalpy-porosity scheme

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ABSTRACT

This study investigates the effects of consistently altering the PCM thermophysical properties, the porosity constant, and their combination on the accuracy of melting models. To the best of the authors' knowledge, this is the first systematic evaluation of this coupled effect, which challenges a common literature practice: to validate the model and use it with another PCM or in a completely different scenario. To this end, a mathematical model is formulated and numerically solved with Ansys® Fluent, being validated with two experimental datasets, comprising distinct geometries and PCM properties. Initially, the discussion centers on the alteration of the porosity constant, revealing its sensitivity in both isothermal and non-isothermal PCMs, with a more pronounced influence in materials exhibiting a mushy region. Subsequently, the effect of altering the parameters included in the Boussinesq approximation was assessed, being shown that a proper selection of the coefficient of thermal expansion is more critical than that of density, altering the liquid fraction in 7.5% and 0.5%, respectively. The phase-change temperature also influenced the results, with a 6.9% higher deviation when switching from the liquidus to the average solidus–liquidus temperature. Furthermore, it was evidenced that, by holding the porosity constant of validation in another geometry, deviations as high as 38.0% could be achieved, whereas this error could be reduced to 5.7% with a proper adjustment. This discrepancy may be even greater if the PCM is altered, leading to unclear results. Lastly, the study provides valuable insight suggesting that, as natural convection intensifies, reducing the porosity constant can mitigate errors. These findings hold practical significance for models developed using the enthalpy-porosity scheme, emphasizing the importance of careful consideration when extending the model applicability beyond the validation conditions.

1. Introduction

The incorporation of thermal energy storage (TES) with renewable energy is a promising alternative towards a more sustainable economy [1]. Latent thermal energy storage (LTES), particularly, has been investigated in several energy-related fields, including concentrated solar power plants [2], photovoltaic [3] and photovoltaic/thermal collectors [4], solar heating [5], heat pumps [6], refrigeration appliances [7], and thermal management of buildings [8]. Novel applications of phase-changes materials (PCMs) are also reported [9]. Compared to sensible thermal energy storage, the main advantages of LTES are the higher storage density and near isothermal charge/discharge process, making PCMs suitable for large-scale and small-scale utilization. In contrast, these materials present low thermal conductivities, which pose a strong barrier to the widespread utilization of LTES.

To overcome the above-mentioned limitation and enhance the response of LTES units, several types of research have been performed. Traditional techniques include the utilization of metal/carbon-based foams, fins, and nano-enhanced PCMs (NePCMs), consisting of a base PCM containing nano-particles with high thermal conductivity. Concerning extended surfaces in shell-and-tube heat exchangers, several arrangements are being proposed, e.g. axial [10], radial [11], helical [12], and deformable [13] fins. Another proven approach for a better performance is the utilization of multiple (cascaded) PCMs [14]. Furthermore, other techniques that can be classified as non-conventional are also attracting the attention of several researchers, such as melting/solidification assisted by magnetic fields [15] and rotating systems [16]. A comprehensive discussion in these and in other topics is provided in ref. [17]. A common characteristic of such studies is to validate a mathematical model and explore it in different conditions to find the best scenarios. Nevertheless, it frequently happens that

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Nomenclature

C	porosity constant [Pa s/m^{-2}]
c_p	specific heat at constant pressure [J kg^{-1} K^{-1}]
f	liquid fraction [–]
g	gravitational acceleration [m/s^{-2}]
h	sensible enthalpy [J kg^{-1}]
H	specific (total) enthalpy [J kg^{-1}]
k	thermal conductivity [W m^{-1} K^{-1}]
L	latent heat of fusion [J kg^{-1}]
Nu	Nusselt number [–]
P	pressure [Pa]
Ra	Rayleigh number [–]
r	radial coordinate [m]
S_u	source term [Pa/m]
t	time [s]
T	temperature [°C]
u	velocity vector [m/s]
x	x coordinate [m]
y	y coordinate [m]
X	characteristic length [m]
ΔH	specific latent heat [J kg^{-1}]

Greek symbols

α	thermal diffusivity [m^2/s]
β	thermal expansion coefficient [K^{-1}]
δ	liquid layer thickness
μ	dynamic viscosity [Pa s]]
ν	kinematic viscosity [m^2/s]
ρ	density [kg^1 m^{-3}]

Subscripts

0	initial
l	liquid
m	melting
ref	reference
s	solid
W	wall

Abbreviations

AAD	absolute average deviation
AHC	apparent heat capacity
CFD	computational fluid dynamics
CGI	grid convergence index
ITA	iterative time advancement
LTES	latent thermal energy storage
NePCM	nano-enhanced phase-change material
NITA	non-iterative time advancement
PCM	phase-change material
TES	thermal energy storage

validation is carried out with a different PCM or in a completely different condition than that in which the model is to be used, probably due to the absence of available experiments. To illustrate the above-mentioned validation problem, Table 1 lists some numerical works.

Due to the complex nature of melting/solidification processes, phase-change models require a set of assumptions to make the simulation a viable task. Even doing so, the computational resources required to perform such works are significant and the processing can be extended from a few days to several weeks, depending on the problem. In this sense, several researchers are engaged on findings alternatives to reduce the computational time without losing the model accuracy. Ezzat and Abdel-Rehim [27] compared the usage of iterative time advancement (ITA) and non-iterative time advancement (NITA) schemes to simulate the melting of lauric acid. The authors solved several cases with both methods and reported that NITA achieved a 96.4 % reduction on the simulation time, while the maximum difference for the melting time was 0.27 %, compared to NITA. Demirkıran and Cetkin [28] evaluated the performance of a melting model using constant and adaptive time step advancements. For the considered conditions, the adaptive approach reduced the computational time in 72 %, with a maximum liquid fraction difference of 1.6 %.

All the investigations listed in the last paragraph and the majority of models found in the literature are established on CFD codes that adopt the enthalpy-porosity approach [29]. In this scheme, the energy equation is written in an enthalpy formulation, while a source term is added at the momentum equation to handle the phase-change (mushy) region, considered a porous medium. Besides porosity, described by the liquid fraction, the source term also includes the porosity constant (C), being the recommended values from 10^4 to 10^7 [30], although higher values have also been reported. Kheirabadi and Groulx [31] reported the dependency between this constant and the melting temperature range, recommending the verification with other material properties. Nevertheless, the boundary conditions and the geometry also represent important factors in this discussion. In practice, after establishing all PCM properties, verifying grid and time step independency results, and other numerical parameters, the porosity constant is used as an "adjustment factor" to find the best accuracy, when simulation results are contrasted with experiments. It is quite common to find studies adopting $C = 10^5$ as a default value, without testing its influence on the results. Nevertheless, discussions on the proper value of C are available in the literature and investigations reporting its variation are becoming more frequent. As it could be expected, the best C value varies for different PCMs, as verified by contrasting the references [32] ($C = 10^8$ for RT27) and [33] ($C = 10^9$ for sodium nitrate), although the value of the porosity constant cannot be considered unique for a given PCM. Therefore, if the model is validated with another PCM, holding the porosity constant unaltered, significant deviations may be incurred in. Besides, large divergences on the best C value can also be noted for the same PCM, e.g. lauric acid, by comparing some studies in Table 2, although different assumptions were considered.

Still in Table 2, ref. [34] reported variations on the most suitable value by just turning the system from vertical to horizontal oriented, while ref. [36] found different values when changing the wall temperature. This finding indicates that, if the material properties and the other parameters are unaltered, the root cause for this behavior would be the intensity alteration of natural convection. Tian et al. [37] also reported different C values when investigating the melting of lauric acid in non-finned (10^6) and finned (3×10^6) vertical enclosures. Therefore, the aforementioned behaviors suggest that there is a dependency between the porosity constant and the intensity of natural convection. This implies that the new condition under investigation by the simulation model should not present strong differences on the PCM motion intensity, compared to the validation set. This necessity to adjust the porosity constant for new scenarios can be considered a limitation of the enthalpy-porosity method. In this regards, Reichl et al. [38] performed a comparison between two models to solve melting and solidification problems, both implemented in Fluent software. The first used the enthalpy-porosity scheme by turning on the melting/solidification mode. The second was developed considering the apparent heat capacity (AHC) method (melting/solidification mode off), in which the phase-change is included by modifying the specific heat curve to include the latent heat of melting/solidification. In this case, the porosity constant, included on the source term of the momentum equation, is not needed, but a very high viscosity value must be inputted to the solid PCM, in

Table 1
A sample of numerical studies employing different systems and/or PCMs that those of validation.

Reference	System (model)	PCM (model)	System (validation)	PCM (validation)
[18]	Vertical shell-and-tube with radial fins	RT55	Vertical shell-and-tube	RT35
[19]	Horizontal shell-and-tube with arrow fins	N-octadecane	Horizontal shell-and-tube	Lauric acid
[20]	Trapezoidal cavity heated by a wavy wall	n-eicosane	Rectangular cavity heated by one side	Gallium
[21]	Finned plate heat exchanger	RT82	Rectangular (vertical) cavity heated by both lateral sides	Hydrated salt ($T_m = 58.4$ °C)
[22]	Printed circuit heat exchanger with cascaded PCMs	3 PCMs enhanced with expanded graphite or 5 PCMs without it (180 °C < T_m < 318 °C)	Parallel slabs heat exchanger	Paraffin ($T_m = 21.7 = °C$)
[23]	Horizontal shell-and-tube (double pass) with NePCMs	Capric acid enhanced with Copper and graphene oxide particles	Rectangular cavity heated by one side	Lead (Pb)
[24]	Pear-shaped container with fins	N-octadecane enhanced with Al_2O_3 nanoparticles	Square cavity heated by one side (numerical)	Paraffin ($T_m \approx 47°C$) enhanced with Al_2O_3 nanoparticles
[25]	Horizontal shell-and-tube with fins	N-octadecane enhanced with Al_2O_3 nanoparticles	Spherical capsule (constrained melting)	N-octadecane without particles
[26]	Horizontal shell-and-tube with lobulus-shaped tube	Paraffin ($T_m \approx 54°C$) enhanced with copper particles	Square cavity heated by one side (numerical)	Paraffin ($T_m \approx 47°C$) enhanced with Al_2O_3 nanoparticles

Table 2
Sample of papers reporting the alteration of the porosity constant (C) value for lauric acid melting.

Reference	System	C values	Best results
Fadl and Eames [34]	Rectangular cavity	10^5, 5×10^5, 10^6, 10^7 for horizontal and vertical orientations, plus 2×10^5 for horizontal.	For vertical orientation, $C = 5 \times 10^5$ performed best, while for horizontal, 2×10^5 was the best
Oliveski et al. [35]	Rectangular cavity with fin	Non informed	$C = 10^{11}$
Troxler et al. [36]	Inclined rectangular cavity	10^4, 3×10^4, 5×10^4, 7×10^4, 10^5	$C = 10^5$ and 3×10^4 were the best for wall temperatures of 60 °C and 70 °C, respectively.

order to force its velocity approach zero. According to the authors, a continuous viscosity function should be used for the entire temperature range to allow stable results. It was concluded that both methods provide accurate results, but the AHC was recommenced due to its independency with the porosity constant, although an artificial viscosity needs to be used.

It could be suggested that, considering all temperature-dependent properties and accounting for the volumetric expansion of the PCM by the volume of fluid (VOF) model, the dependency of the porosity constant value with the natural convection intensity would not exist. Based on the literature published so far, the authors cannot affirm so, but this idea needs to be verified. However, the VOF model has been seldom adopted due to its significantly higher computational effort. Therefore, the problem of the dependency between the porosity constant and the natural convection intensity still would exist in practice. Another comment concerning Tables 1 and 2 is that several investigations adopt the Boussinesq approximation, Eq. (1), to reduce the simulation time, keeping an acceptable precision, as demonstrated in some investigations, e.g. [38,39]. This benefit can be achieved once the density appearing in all governing equations is assumed constant, while only the buoyance term of the momentum equation considers its variation, although indirectly.

$$\rho = \rho_{\text{ref}}\left[1 - \beta\left(T - T_{\text{ref}}\right)\right] \quad (1)$$

One may be in doubt in what value to consider as reference for density. Some authors argue that, considering the stored energy, the mean density between the solid and liquid PCM is a more suitable choice. On the other hand, from a hydrodynamics point of view, the density value at the liquid state could be the most suitable option, once the melting process is dominated by natural convection [39]. Recently, Troxler et al. [36] recommended to use different values of density for the liquid and solid PCM. However, this choice lead to longer simulation times. The other PCM properties should also be selected with caution, since they can directly affect the results accuracy.

Fatty acids, such as lauric acid, have become widely investigated PCMs, due to their characteristics of high heat capacity, low vapor pressure, and good long-term thermochemical stability [40]. Besides, its melting temperature makes it suitable for a wide range of applications involving heating and cooling, when the desired temperature is about 45 °C. In this sense, this material was selected for the present investigation. Table 3 compiles the thermophysical properties of the lauric acid, for different sources.

Although the claimed purity of lauric acid used in the works of Table 3 is, generally, 99 %, there is a significant variation in its reported properties. Therefore, a proper combination of the C constant with the thermophysical properties must be done. The adequacy of this choice needs to be confirmed by comparing simulating and experimental results, considering similar conditions and the same PCM.

To the best of the authors' knowledge, a comprehensive study evaluating the impact of altering the thermophysical properties of PCMs in a physically meaningful manner remains absent in the existing literature. Many references merely report random variations [49], without delving into the systematic effects of adjusting these properties. Moreover, while Troxler et al. [36] made an attempt to combine density with porosity constant, Table 3 illustrates that various other properties also exhibit significant disparities across different references. Furthermore, the synergy between selecting appropriate PCM properties and setting the porosity constant value remains a relatively unexplored territory. Lastly, there is a scarcity of research addressing the implications of validating a model for a specific configuration and subsequently employing it in different geometries, while maintaining the original set of variables. This gap is especially pertinent as significant errors can potentially arise when utilizing a simulation model under conditions differing from those used for validation. Therefore, the present investigation seeks to bridge these aforementioned research gaps, with particular emphasis on the latter, which merits thorough discussion given its potential to impact the accuracy and applicability of the simulation model in diverse scenarios. To this end, a mathematical model is developed and numerically solved, employing Fluent software, to examine the melting behavior of lauric acid. The accuracy of the model is rigorously validated against experimental data collected from prior studies, encompassing both a rectangular enclosure and a shell-

Table 3
Thermophysical properties of lauric acid found in different references.

Reference	Phase-change temperature [°C]	Latent heat [kJ/kg]	Specific heat [J/(kg.K)]	Density [kg/m^3]	Thermal conductivity [W/(m.K)]	Viscosity [Pa.s]	Coefficient of thermal expansion [1/K]
Mahdi et al. [41]	44.5 (solidus); 48.3 (liquidus).	178.78	2200	857	0.157	0.0067	0.0006
Yuan et al. [42]	44.2 (melting)	173.8	2300 (liquid)	862.9 (60 °C), 856 (70 °C), 848.3 (80 °C),	0.147 (liquid)	0.005336 (60 °C), 0.004269 (70 °C), 0.003469 (80 °C).	0.000615
Cao et al. [43]	44.2	174.9 (melting); 173.4 (solidification)	2300 (liquid)	862.9 (solid); 856 (liquid at 70 °C).	0.147 (liquid)	–	0.000615
Kalapala and Devanuri [44]	43.5 (solidus); 50.0 (liquidus).	156.8	1390 (solid); 1570 (liquid).	–	–	–	–
Shokouhmand and Kamkari [45]	43.5 (solidus); 48.2 (liquidus).	187.2	2180 (solid); 2390 (liquid).	940 (solid); 885 (liquid).	0.16 (solid); 0.14 (liquid).	–	–
Liu and Groulx [46]	44	182	2400 (solid); 2000 (liquid).	950 (solid at 20 °C); 873 (liquid at 45 °C).	0.150 (solid); 0.148 (liquid).	0.008 (at $\sim T_m$)	–
Shaker et al. [47]	40.5 (solidus); 42.5 (liquidus).	129.5	2180 (solid); 2390 (liquid).	940 (solid); 885 (liquid).	0.16 (solid); 0.14 (liquid).	0.005930	0.00083
Haotian et al. [48]	43 (solidus); 44 (liquidus).	143.8	2300	848.3	0.147	0.003469	0.000615

and-tube unit as benchmark cases. After validation, a systematic exploration of various factors is undertaken. Initially, an isolated assessment is conducted to identify the influence of the porosity constant and key thermophysical properties of the PCM. Subsequently, an analysis is carried out to quantitatively investigate the interplay between the C constant and the material properties, elucidating the consequences of altering the system geometry. Furthermore, this research contributes to a deeper understanding of the relationship between the porosity constant and the intensity of natural convection, when using the enthalpy-porosity scheme.

2. Numerical methodology

2.1. Physical domain

In this study, two distinct reference configurations are employed. The first configuration involves a rectangular cavity measuring 120 mm in length and 50 mm in height, as depicted in Fig. 1a. In this setup, the bottom wall is maintained at a constant temperature of 70 °C, while the remaining walls are treated as perfectly insulated, ensuring adiabatic conditions. The initial temperature of the PCM is set at 25 °C. This particular configuration corresponds to the scenario with an inclination angle of 0°, as established by Kamkari et al. [50]. The second reference configuration represents a shell-and-tube unit, characterized by an inner tube with an outer diameter of 40 mm and an outer tube with an inner diameter of 80 mm, as illustrated in Fig. 1b. In this configuration, the inner tube serves as the heated wall, maintained at a temperature of 80 °C, while the shell is assumed to be adiabatic. The initial temperature of the PCM in this case is set at 20 °C, in accordance with the experimental setup by Yuan et al. [42]. Throughout this study, lauric acid is adopted as the PCM, and its thermophysical properties for the reference cases 1 and 2 are detailed in Table 4. It is worth noting that, the properties from ref. [50] were originally derived from a prior work of the authors, ref. [45], which includes distinct values for the liquid and solid phases of the PCM. However, specific information regarding viscosity and the coefficient of thermal expansion was not provided. For case 1, it was assumed the specific heat, density, and thermal conductivity as the average values between the liquid and solid phases, while the values for viscosity and the coefficient of thermal expansion were obtained from ref. [42].

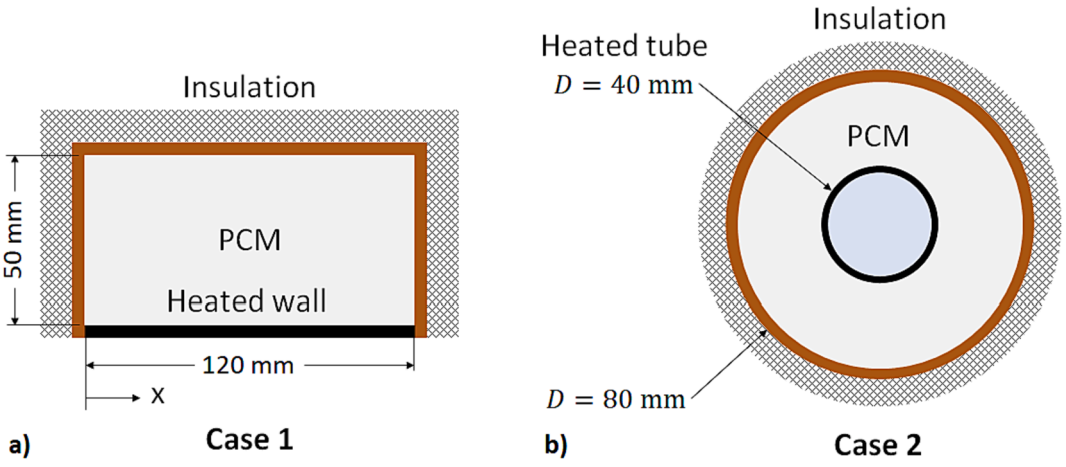


Fig. 1. Physical domains under investigation: a) Case 1 – rectangular cavity; b) Case 2 – shell-and-tube unit.

Table 4
Thermophysical properties of lauric acid for cases 1 and 2.

Reference	Phase-change temperature [°C]	Latent heat [kJ/kg]	Specific heat [J/(kg.K)]	Density [kg/m^3]	Thermal conductivity [W/(m.K)]	Viscosity [Pa.s]	Coefficient of thermal expansion [1/K]
Case 1 [45]	43.5–48.2	187.2	2285*	912.5*	0.15*	0.005336**	0.000615**
Case 2 [42]	43.5***	173.8	2300	862.9	0.147	0.003469	0.000615

* Average of solid and liquid values.
** From ref. [42].
***The value of ref. [42] is 44.2. Nevertheless, 43.5 demonstrated to be a best choice, as discussed in section 3.2.

2.2. Mathematical model

The model of the PCM melting is based on the enthalpy-porosity scheme [29], which solves the mass, momentum, and energy conservation equations. Furthermore, the present model considers the following assumptions: (i) the liquid PCM behaves as a Newtonian fluid, (ii) the flow of the Liquid PCM is laminar and incompressible, (iii) thermophysical properties are constant, (iv) contact resistance between the solid PCM and tube wall is negligible, (v) heat conduction through the axial direction of the tubes is negligible, and (vi) Boussinesq approximation (Eq. (1) is valid, i.e. the volumetric thermal expansion is neglected. Therefore, the set of governing equations are written as Eqs. 2–4.

$$\nabla\cdot\vec{u} = 0 \tag{2}$$

$$\frac{\partial}{\partial t}(\rho\vec{u}) + \nabla\cdot(\rho\vec{u}\vec{u}) = \rho\vec{g}\beta\left(T - T_{\text{ref}}\right) - \nabla P + \mu\nabla^2\vec{u} + \vec{S}_{\text{u}} \tag{3}$$

$$\frac{\partial(\rho H)}{\partial t} + \nabla\cdot(\rho H\vec{u}) = \nabla\cdot(k\nabla T) \tag{4}$$

where $\vec{u}$ is the velocity vector, g is the gravitational acceleration, P is pressure, μ is the dynamic viscosity, and $\vec{S}_{\text{u}}$ is a momentum source term that deals with the PCM velocity as it solidifies or melts. The same equations, written in cylindrical coordinates, apply for the shell-and-tube unit, but are not presented here for brevity. The momentum source term is given by Eq. 5.

$$\vec{S}_{\text{u}} = -A\vec{u} \tag{5a}$$

$$A = C\frac{(1-f)^2}{f^3 + q} \tag{5b}$$

where C is the porosity constant, presented in the Introduction Section, and q is a small number (0.001 – default of Fluent) to avoid division by zero. Concerning the energy equation, the total enthalpy (H) is split in sensible (h) and latent heat (ΔH) terms, as indicated by Eq. 6.

$$H = h + \Delta H \tag{6a}$$

$$h = h_{\text{ref}} + \int_{T_{\text{ref}}}^{T} c_{\text{p}}dT \tag{6b}$$

$$\Delta H = fL \tag{6c}$$

where c_{p} is the specific heat at constant pressure and L is the latent heat of fusion. A linear relation between enthalpy and temperature is assumed at the mushy region, such that the liquid fraction can be recovered from the temperature field, and vice-versa, according to Eq. (7).

$$f = \begin{cases} 0, if T < T_{\text{s}} \\ \dfrac{T - T_{\text{s}}}{T_{\text{l}} - T_{\text{s}}}, if T_{\text{s}} \le T \le T_{\text{l}} \\ 1, if T > T_{\text{l}} \end{cases} \tag{7}$$

where T_{s} and T_{l} are the solidus and liquidus temperatures, respectively. Besides, the liquid fraction is updated iteratively from the enthalpy field (energy equation- Eq. (4)), according to Eq. (8).

$$f^{k+1} = f^k + \frac{A_{\text{P}}h_{\text{P}}}{\rho L} \tag{8}$$

where $k+1$ denotes the next iteration, k is the current iteration, A_{P} is the local coefficient of the discretized momentum equation, and h_{P} is the local sensible enthalpy. The initial and boundary conditions for this work are as follows:

- The no-slip condition imply in zero velocity at the walls, i.e. $\vec{u} = 0$;
- At the heated walls, the temperature assumes the prescribed wall temperature, i.e.$T = T_{\text{W}}$;
- At the insulated walls, adiabatic conditions apply, i.e. $\frac{\partial T}{\partial x} = \frac{\partial T}{\partial y} = \frac{\partial T}{\partial r} = 0$;
- The initial condition is a uniformly distributed temperature, i.e. at $t = 0$, $T = T_0$.

2.3. Numerical implementation

The mathematical model was discretized using the finite volume method and solved via ANSYS® Fluent 17. In this solution, a pressure-based solver was employed with PISO algorithm for pressure–velocity coupling. The interpolation function for the diffusive terms in the energy equation utilized second-order central differences, consistent with Fluent default settings. For the interpolation of convective terms in both the momentum and energy equations, a second-order upwind scheme was employed. The discretization of pressure and time was carried out using PRESTO! and a first-order implicit scheme, respectively. Regarding the under-relaxation factors, the following values were selected: 0.3 for pressure, 0.7 for momentum, 0.9 for liquid fraction updates, and 1 for energy. The iterative process for each time step was terminated when the residuals for the continuity, momentum, and energy equations reached values below 10^{-3}, 10^{-3}, and 10^{-6}, respectively, although, continuity was found to be the constraint. An schematic flowchart of the numerical procedure to solve the mathematical model is illustrated in Fig. 2. It does not include the geometry definition nor the mesh generation, which are done previously.

Since both cases are symmetric, only half of the domain could be solved. However, the experiments of Kamkari et al. [50] showed a non-symmetric pattern of the melted regions in some time intervals. Therefore, it was decided to simulate the whole domain in case 1. For the shell-and-tube unit (case 2), only half of the domain was simulated. This choice was based on the experimental visualization reported by Cao et al. [43], who performed investigations on the melt of lauric acid in a shell-and-tube unit.

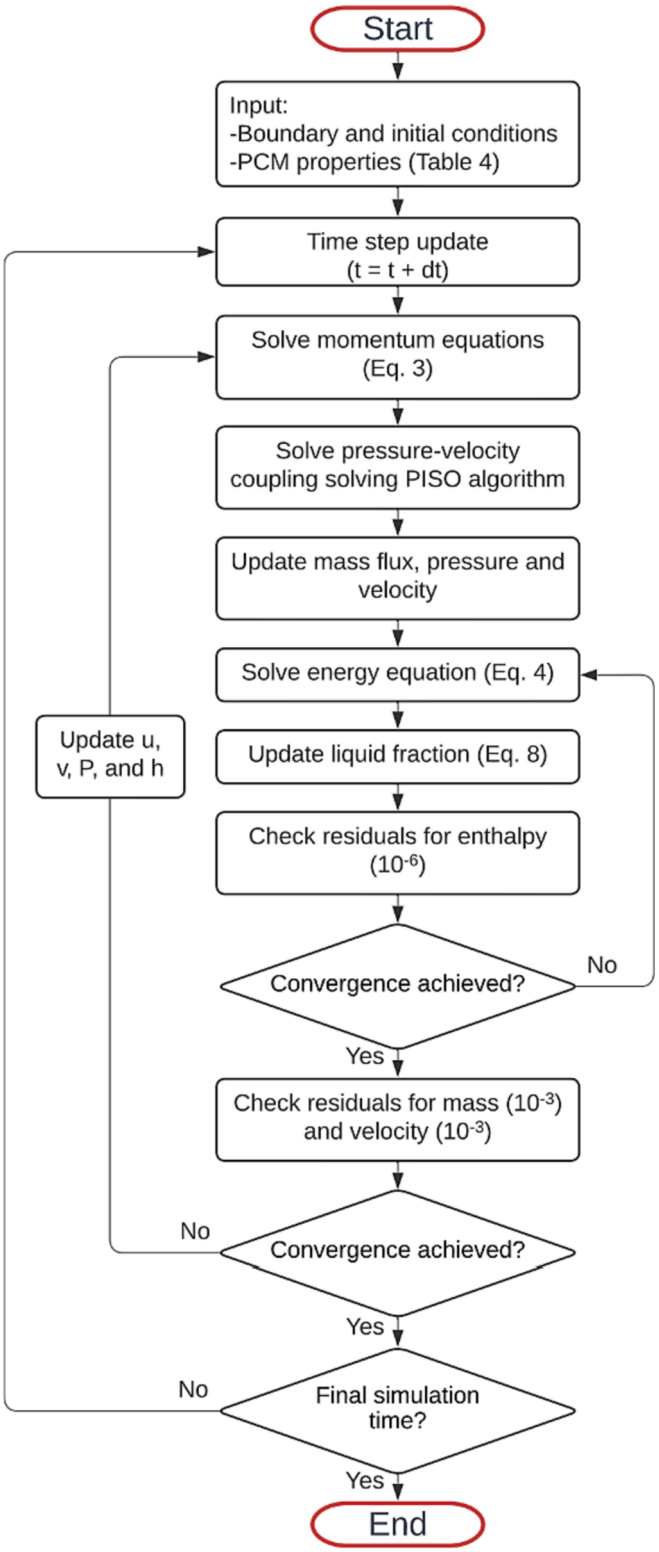


Fig. 2. Schematic flowchart of the numerical procedure to solve the mathematical model.

2.4. Mesh and time step verification

For both cases, structured grids with quadrilateral elements were adopted. The mesh sensitivity analysis was carried out with the grid convergence index (GCI) method [51], which represents the uncertainty due to the numerical discretization. It is expressed by Eq. 9.

$$GCI = \frac{1.25 e_a}{r_{21}{}^{p} - 1} \tag{9a}$$

$$e_a = \left| \frac{\varphi_1 - \varphi_2}{\varphi_1} \right| \tag{9b}$$

$$p = \frac{|\ln|\varepsilon_{32}/\varepsilon_{21}| + q|}{\ln(r_{21})} \tag{9c}$$

$$q = \ln\left(\frac{r_{21}{}^{p} - s}{r_{32}{}^{p} - s} \right) \tag{9d}$$

$$s = 1 \bullet sgn(\varepsilon_{32}/\varepsilon_{21}) \tag{9e}$$

where $\varepsilon_{32} = \varphi_3 - \varphi_2$, $\varepsilon_{21} = \varphi_2 - \varphi_1$, φ_k is the solution (liquid fraction) on the kth grid, and r is the grid refinement ratio. The function *sgn* on Eq. (9e) assumes the values of −1, 0, or 1. For case 1, three meshes with 9,261, 16,600, and 29,304 volumes were considered. The GCI was calculated in six different periods (900, 1800, 2700, 3600, 4500, and 5400 s) of the total solution time, with an average value of 2.35 %, considering the second grid. For case 2, meshes with 19,229, 35,490, and 62,315 volumes were chosen and five periods were selected (400, 800, 1200, 1600, and 2000 s) to calculate the GCI, with an average value of 3.87 %, for the second grid. Based on the results of Fig. 3a and Fig. 4a and considering average GCI of 2.35 % and 3.87 %, meshes with 16,600 and 35,490 volumes were selected for cases 1 and 2, respectively.

A time step verification was also performed, with values of 0.2 s, 0.1 s, and 0.05 s for both cases, and the trend of the liquid fraction is presented in Fig. 3b and 4b, for cases 1 and 2, respectively. As a good trade-off between accuracy and computational time, 0.1 s were selected for both cases. It is worth noting that, for case 2, 0.2 s also provided an independent solution for the time step. Nevertheless, the number of iterations to achieve convergence significantly increased, such that the simulation time was not advantageous, compared to that of 0.1 s.

2.5. Validation

To assure the adequacy of the proposed model, the predicted results for the liquid fraction (Fig. 5a) and the average convective coefficient (Fig. 5b), for case 1, are contrasted with the experimental values reported by Kamkari et al. [50]. The heat transfer coefficient ($\bar{h}$) was calculated with Eq. (10) [30]. Concerning the liquid fraction, a good agreement was found, with maximum and absolute average deviations (AAD) of 11.2 % (at 800 s) and 3.3 %, respectively. For the convective coefficient, larger deviations were observed, being the maximum and AAD values of 41.5 % (at 500 s) and 9.1 %, respectively. Nevertheless, the trend is consistent and the maximum deviation occurred at the beginning of the melting process. From 1000 s onwards, the maximum deviation was 11.6 %. Therefore, this validation set, with the porosity constant equal to 7.5×10^5, was considered satisfactory to validate the model. The experimental ranges in case 1 were considered to be between 400 s and 5300 s and between 400 s and 5100 s for the liquid fraction and convective coefficient, respectively, in order to avoid data extrapolation. The absolute average deviation was calculated with Eq. (11).

$$\bar{h} = \frac{q''}{T_w - T_m} \tag{10}$$

$$\text{AAD} = 100\% \left(\frac{1}{n} \sum_{n} \left| \frac{\text{predicted} - \text{experimental}}{\text{experimental}} \right| \right) \tag{11}$$

where q'' is the heat flux at the hot wall and n is the number of calculation points. For case 2, the liquid fraction predictions were contrasted with the experiments of Yuan et al. [42], as presented in Fig. 6. Maximum and AAD of 9.3 % (at 550 s) and 4.51 % were found for the experimental range, between 550 s and 1700 s. The predictions are also

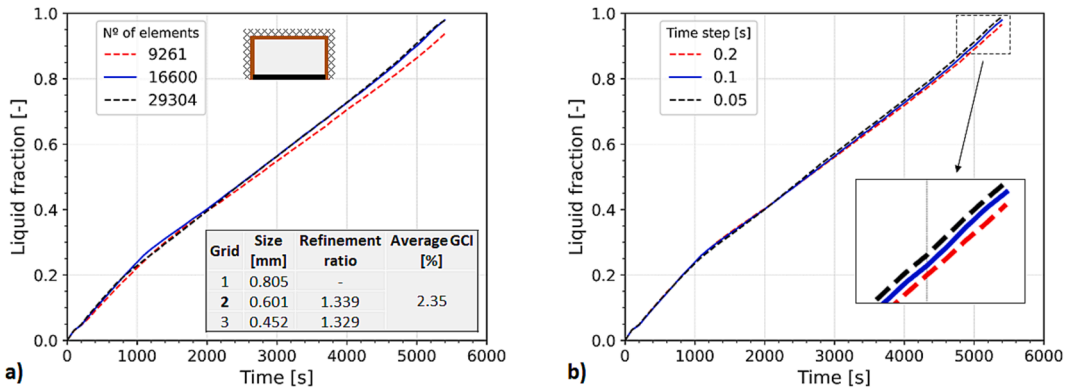


Fig. 3. A) mesh and b) time step verification results for case 1.

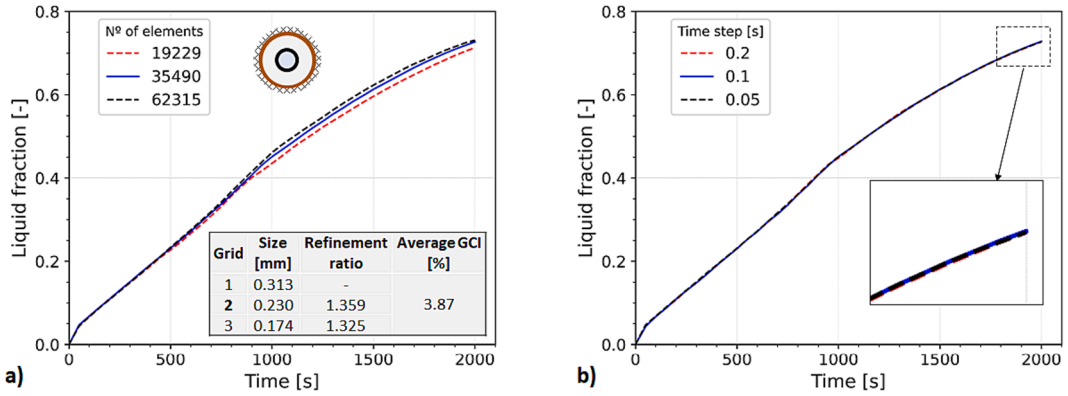


Fig. 4. A) mesh and b) time step verification results for case 2.

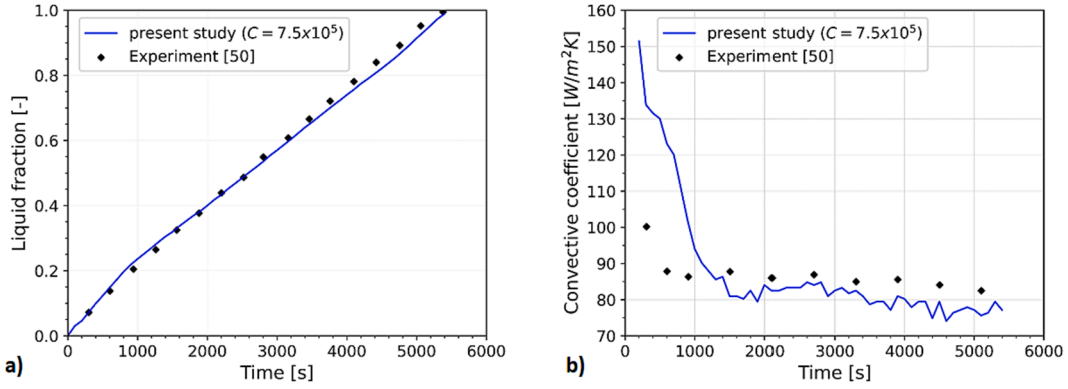


Fig. 5. Comparison of the predicted: a) liquid fraction; b) convective coefficient against the experimental values of Kamkari et al. [50], for case 1.

consistent with the numerical results reported by Haotian et al. [48]. In this sense, this second validation set, with $C = 7.5 \times 10^4$, was also considered satisfactory, for case 2. These values of the porosity constant chosen for validation were those who led to the best accuracy, as presented in the next section (3.1).

3. Results and discussions

3.1. Porosity constant

Initially, the isolated effect of the porosity constant on the results accuracy was assessed in terms of the liquid fraction. For case 1 (Fig. 7a), five values between 2.5×10^5 and 10^6 were tested, being $C = 7.5 \times 10^5$ the best. A typical value of $C = 10^5$ was also tried, but the solution started not converging from about 2000 s onwards. For $C = 2.5 \times 10^5$, convergence was not achieved for several time steps between 4700 s and 5400 s. The maximum and AADs, compared to the experimental data, are indicated in Table 5. As observed, despite the lower variation of the C values, AAD of up to 11.8 % were identified. For case 2, eight values between 10^4 and 10^7 were verified, as shown in Table 5. For a better visualization, some of them were omitted in Fig. 7b. As noted, the value of 10^4 over predicted the liquid fraction up to about 1750 s, when, from there on, it became to under predict it. This behavior could be connected to an alteration of the natural convection intensity at $f \approx 0.4$, as discussed in Section 3.4. For lower values than the reference (7.5×10^4), the behavior was the opposite, i.e. the solution initially under predicted the liquid fraction and at the end of the simulation, started to over predict it.

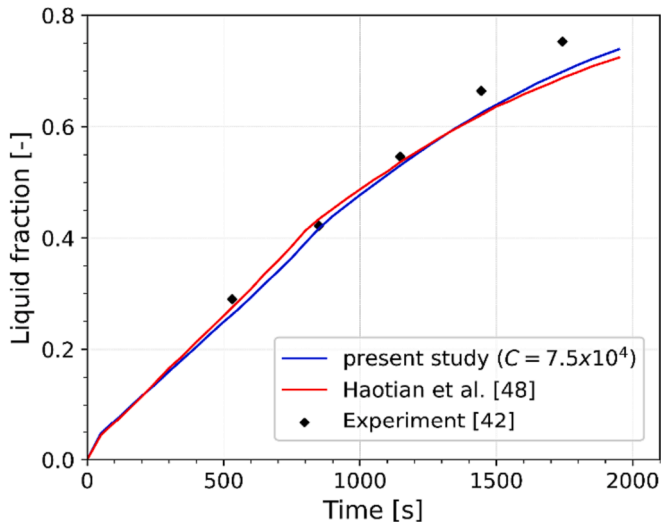


Fig. 6. Comparison of the predicted liquid fraction versus the numerical data of Haotian et al. [48] and the experiments of Yuan et al. [42], for case 2.

Another aspect that could be observed analyzing Table 5 is that case 2 (isothermal) was less impacted by the alteration of the porosity constant, compared to case 1 (non-isothermal). This result agrees with the hypothesis of Vogel and Thess [39]. Nevertheless, it is clear that the isothermal material (case 2) was also affected by the porosity constant. Therefore, based on the present results, it is recommended to check this parameter when validating the model, regardless if a mushy or isothermal material is considered.

3.2. PCM thermosphysical properties

Since Boussinesq approximation is frequently adopted in phase-change simulations, the effect of altering the density and the coefficient of thermal expansion, appearing in Eq. (1), was investigated. The liquid fraction progress over the simulation time is represented in Fig. 8a, for case 1. As it can be observed, a slightly improvement was found when altering from the average (912.5 kg/m^3) to the liquid PCM density (885 kg/m^3, ref. [45], Table 3), where the AAD changed from 5.9 % to 5.4 %, respectively. Nevertheless, this is negligible compared to the influence of β, when the AAD changed from 5.9 % ($\beta = 0.000615\,K^{-1}$) to 13.4 % ($\beta = 0.00083\,K^{-1}$, ref. [47]). For case 2, the density alteration from 862.9 kg/m^3 (at 60 °C, ref. [42]) to 848.3 kg/m^3 (at 80 °C, ref. [42]) was also verified, but the difference on the liquid fraction evolution was negligible. Therefore, selecting appropriate values of β is more critical than the density value.

Considering case 2 (isothermal), the phase-change temperature also produced different results, as depicted in Fig. 8b. The liquidus temperature of case 1 (43.5 °C) demonstrated to be the most representative value to be assumed for the melting temperature, followed by the value of ref. [42] (44.2 °C), and the average of the liquidus and solidus temperatures of case 1 (45.85 °C). In this order, the AADs were 4.8 %, 6.9 %, and 11.7 %, compared to the experimental results. Nevertheless, for solidification, it is suggested to also test the solidus temperature as the phase-change temperature.

3.3. Combination of porosity constant and PCM properties

A practice frequently observed in the literature consists in validating the model for a given PCM, but using it to simulate problems that adopt different PCMs or keeping the original PCM in variable geometric configurations, as illustrated in Table 1. Frequently, the porosity constant value used for validation is kept unaltered when using the model. The effect of such an approach is assessed here.

Initially, the PCM properties and the porosity constant of case 2 were combined with the geometry and boundary conditions of case 1, which means a geometry alteration for an already validated set. The liquid fraction evolution is illustrated in Fig. 9a and, as it can be seen, keeping the original value of $C = 7.5 \times 10^4$ implied in a difference for the total melting time of 1400 s (4000 s versus 5400 s), or 26 %, compared to the experimental results of ref. [50]. Up to the total melting time, i.e. 4000 s, the maximum deviation and AAD as large as 57.9 % and 38.0 %, respectively, were obtained. In order to reduce these discrepancies to 23.8 % and 5.7 %, respectively, a correction should be done on the porosity constant, with a new value of $C = 7.5 \times 10^6$.

Subsequently, the opposite was also verified, i.e. the PCM properties and the porosity constant of case 1 were combined with the geometry and boundary conditions of case 2, as visualized in Fig. 9b, where maximum and AAD of 28.1 % and 23.4 %, respectively, were obtained. Nevertheless, in this situation, the solution did not converge for

Table 5
Deviations on the liquid fraction for different values of the porosity constant for cases 1 and 2.

Case 1 (validation 7.5×10^5)			Case 2 (validation 7.5×10^4)		
C value	Maximum [%]	AAD [%]	C value	Maximum [%]	AAD [%]
2.5×10^5*	22.0	11.8	10^4	16.5	7.0
5×10^5	16.1	5.9	5×10^4	8.7	4.9
6.25×10^5	12.9	3.8	10^5	9.3	4.8
10^6	11.1	4.3	5×10^5	11.0	5.2
			10^6	11.3	5.6
			2.5×10^6	12.4	6.4
			10^7	13.7	7.5

*Calculated up to 4700 s.

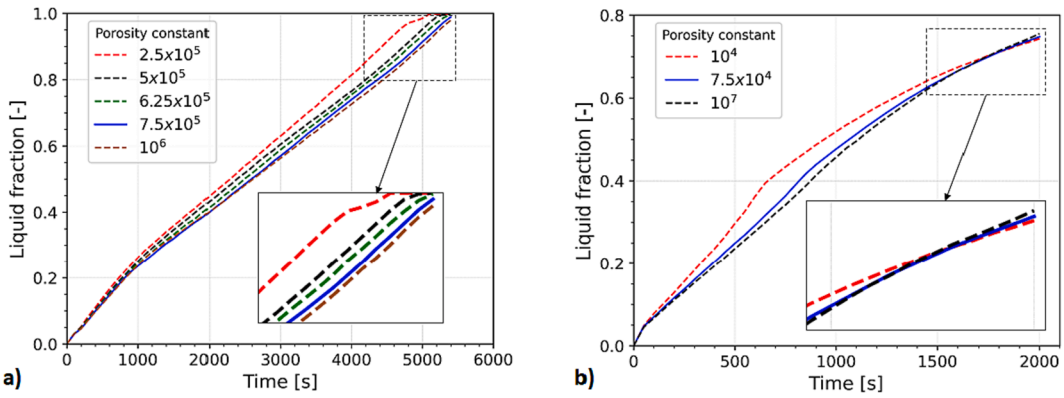


Fig. 7. A. liquid fraction evolution for different values of the porosity constant: a) case 1; b) case 2.

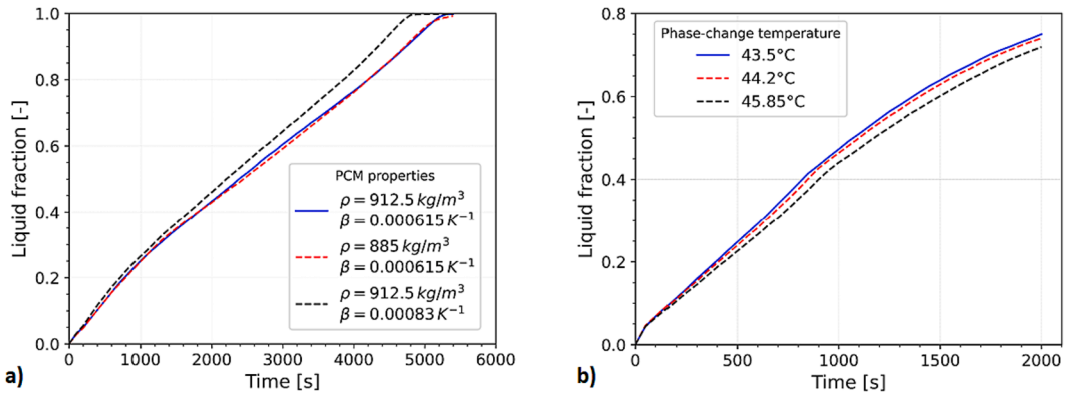


Fig. 8. Liquid fraction evolution by varying the: a) density and coefficient of thermal expansion, in case 1; b) melting temperature, in case 2.

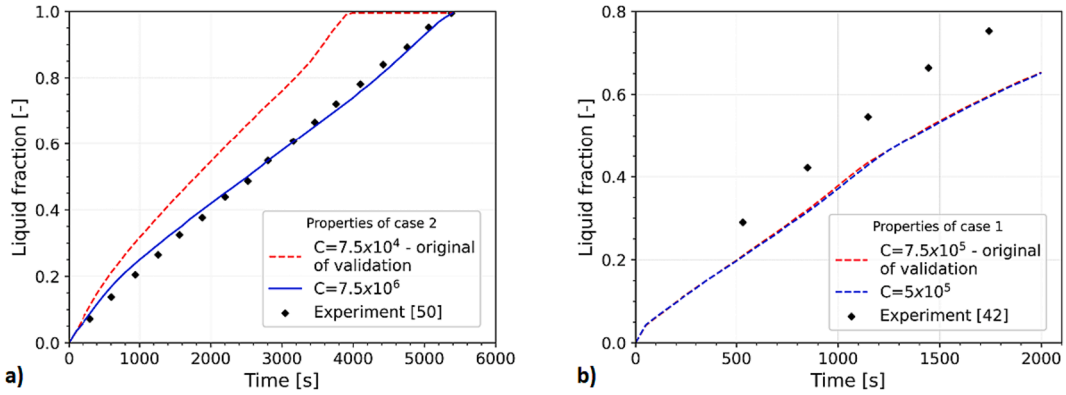


Fig. 9. Liquid fraction evolution for: a) PCM properties of case 2 and geometry of case 1; b) PCM properties of case 1 and geometry of case 2.

$C \leq 5 \times 10^5$, being not possible to assess the necessary correction on the porosity constant that would lead to a satisfactory agreement. These results clearly indicates that the combining effect of the original porosity constant with the new geometry could lead to unreliable findings. It should be highlighted that the PCM properties were not altered, i.e. the same information used in validation was repeated for the new simulation. Therefore, altering the PCM used in novel simulation scenarios, compared to that of validation, could further exacerbate this discrepancy.

3.4. Further insights on the porosity constant and natural convection

As mentioned in the Introduction section, the phenomenon behind the behavior appearing in Fig. 9a is the alteration of the natural convection intensity, which changed the optimal porosity constant from 7.5×10^4 (geometry of case 2) to around 7.5×10^6 (geometry of case 1). Since the Rayleigh number, Eq. (12), depends on the proper choice of a characteristic length, it is not possible to directly compare the natural convection intensity for these two different configurations. Therefore, the velocity and the convective coefficient were examined to verify if any relation is revealed. These variables were contrasted considering instants with the same liquid fraction, since the dimensionless time (Fourier number) also depends on the characteristic length. Noting that the PCM properties are those of case 2, the velocity field is plotted in Fig. 10, for cases 1 and 2, in instants where $f = 0.4$. As it can be noted, the maximum velocity for case 2 (11.8 mm/s) is 80 % higher than that of case 1 (6.56 mm/s), indicating a stronger convective motion for case 2. To verify if this is a trend for the entire melting process, the average velocity of the whole PCM domain is plotted versus the liquid fraction, as shown in Fig. 11a. The maximum liquid fraction for case 2 was 0.75 and, therefore, this threshold was also employed for case 1.

$$\mathrm{Ra} = \frac{g\beta\left(T_{\mathrm{ref},1} - T_{\mathrm{ref},2}\right)X^3}{\alpha\nu} \tag{12}$$

where X is a characteristic length, α is the thermal diffusivity, ν is the kinematic viscosity, and $\left(T_{\mathrm{ref},1} - T_{\mathrm{ref},2}\right)$ is the driving temperature. As it can observed from Fig. 11a, the velocity of case 2 increased up to $f \approx 0.4$, when it started decreasing, differently to what occurred for case 1, where the velocity kept raising. This can be explained by the reduction of natural convection in case 2 and happened because the average temperature of the liquid PCM approached the wall temperature, reducing the buoyancy force that promotes the PCM flow. This is illustrated in Fig. 12 for instants where the liquid fraction is 0.6.

From Fig. 12b, it is possible to see that the upper part of the annulus (geometry of case 2) is almost entirely at the same temperature, approaching the wall temperature of 80 °C, while the remaining liquid phase presents a stratified temperature distribution. In contrast, the geometry of case 1 exhibits a temperature gradient more randomly distributed in the liquid fraction (Fig. 12a), which could justify a higher convective flow. Nevertheless, up to $f \approx 0.4$, the average velocity of case 2 is higher than that of case 1, which can be associated with the natural convection intensity in these situations. To corroborate this hypothesis, the convective coefficient is plotted versus the liquid fraction in Fig. 11b.

According to Jany and Bejan [52], the melting process is governed by four regimes: (i) pure conduction, (ii) mixed conduction and convection (transition), (iii) natural convection, and (iv) shrinking solid. The first is characterized by high initial Nusselt numbers (Nu) that sharply drops to a minimum. In this stage, the melting front is nearly symmetric to the heated wall. In regime (ii) a slightly increase in Nu is observed, which can be attributed to an increase of convection currents. The third regime presents an almost stationary Nu, explained by the dominance of the convection. Lastly, regime (iv) exhibits a decrease of Nu due to the

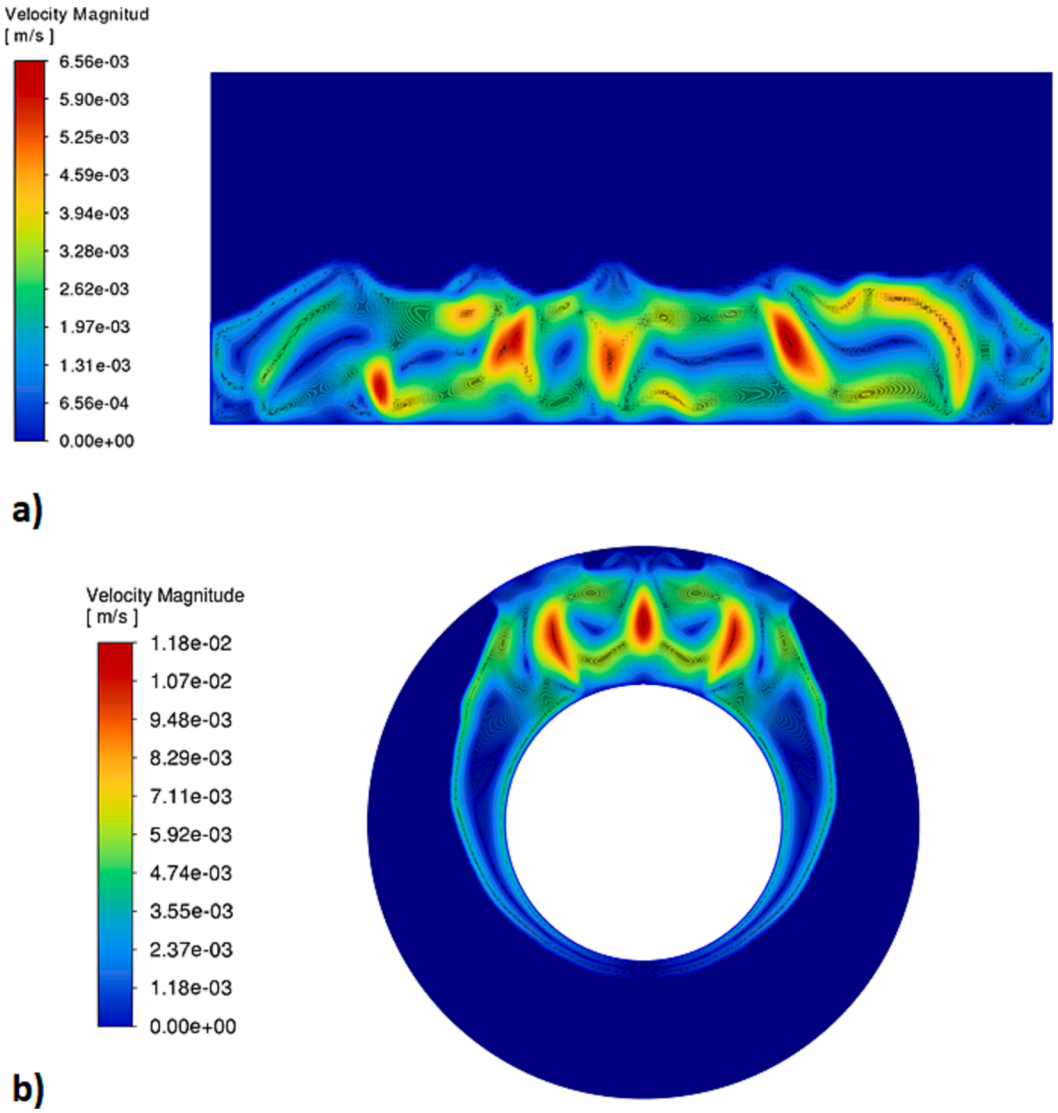


Fig. 10. Velocity field for instants where the liquid fraction is equal to 0.4: a) geometry of case 1; b) geometry of case2.

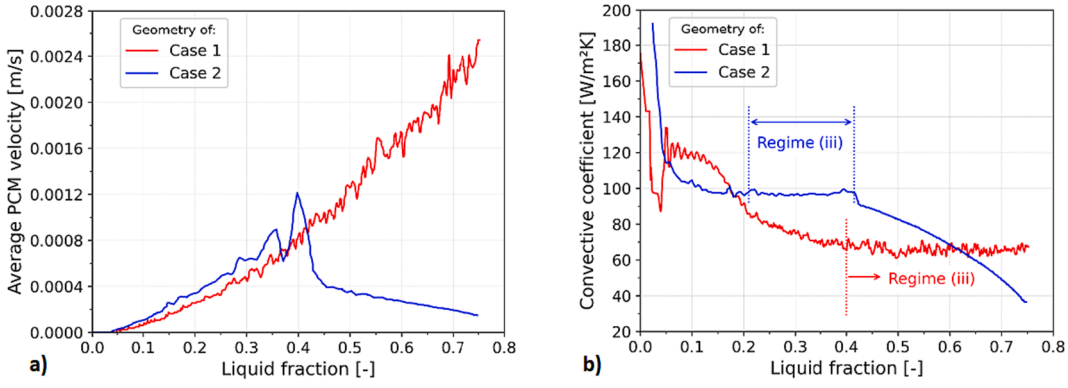


Fig. 11. Evolution of the average: a) PCM velocity; b) convective coefficient for geometries of cases 1 and 2.

shrinkage of the convective flow [45]. In some situations, as in the present study, it is not possible to easily identify the boundaries between these regions, especially the transition one. Nevertheless, the natural convection dominant regime (iii) can be clearly seen in Fig. 11b, for both geometries. The convective heat transfer coefficient is associated to Nu and, for melting/solidification, this nondimensional can be casted in the form of Eq. (13) [53].

$$\overline{\mathrm{Nu}} = \frac{\overline{h}X}{k_l} = cRa_X{}^n\left(\frac{\delta}{X}\right) \tag{13}$$

where h is the convective coefficient, k_l is the thermal conductivity of

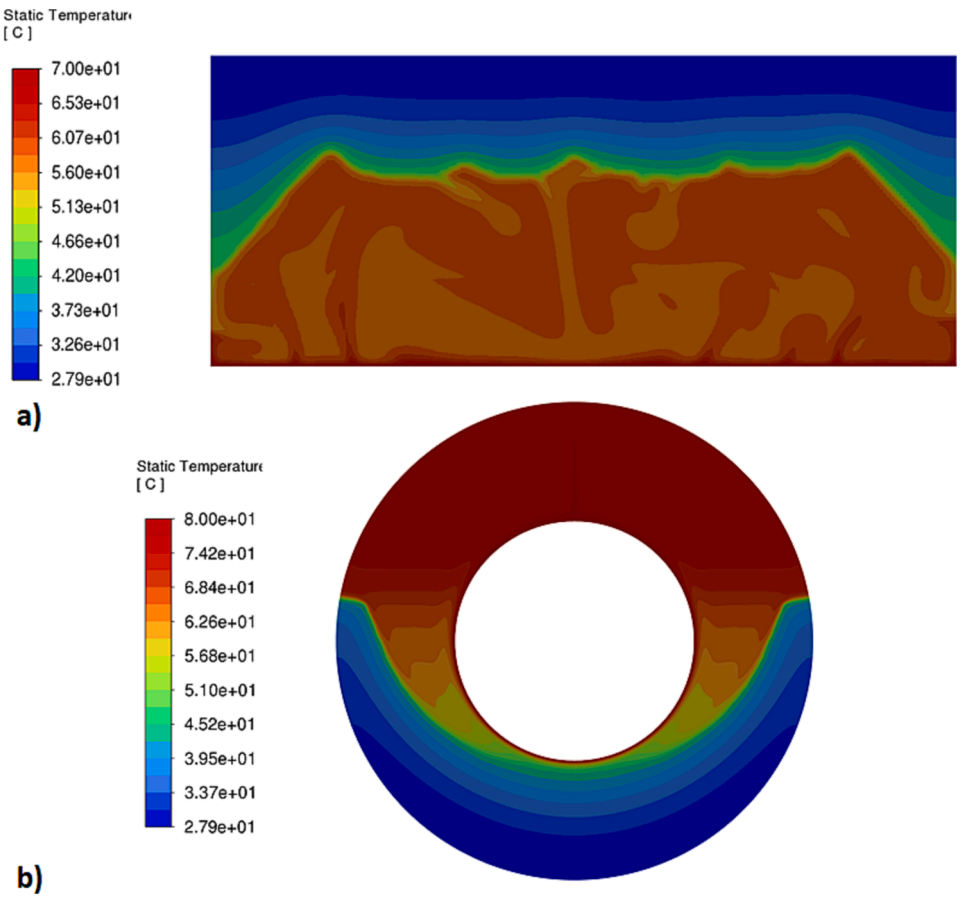


Fig. 12. Temperature field for instants where the liquid fraction is equal to 0.6: a) geometry of case 1; b) geometry of case2.

the liquid PCM, c and n are coefficients derived from experiments, and δ is the thickness of the melted PCM. As observed on Eq. (13), the convective coefficient is proportional to the Rayleigh number, which is an indicator of the natural convection intensity. Hence, considering regime (iii), it can be said that the intensity in case 2 is greater than that of case 1. Concerning the porosity constant, its value was lower for case 2, where natural convection is stronger. This result is also in line with that of Troxler et al. [36], in which the best C value changed from 10^5, for a wall temperature of 60 °C (lower Ra), to 3×10^4, for a wall temperature of 70 °C (higher Ra). Therefore, the lower the natural convection intensity, the higher the porosity constant should be. One attempt to calculate the porosity constant was found in the work of Zeneli et al. [54], who modelled the melting of silicon as the PCM. Although their correlation was related to the viscosity, density, and distance between dendrites secondary arms, the development of other estimators is encouraged.

4. Conclusions

In this work, the role of PCM thermophysical properties, the porosity constant, and the interaction between them were assessed to understand their impact on the accuracy of numerical results. The melting process of lauric acid was evaluated in two different scenarios: a rectangular horizontal enclosure with a non-isothermal (mushy) phase-change (case 1), and a shell-and-tube unit with an isothermal melting (case 2). A mathematical model that adopts the enthalpy-porosity scheme was written and solved with a CFD software, being properly validated with experimental results for both cases. The main outcomes of this investigation are:

- Concerning the porosity constant: for case 1, the best value was $C = 7.5 \times 10^5$. By altering it to 2.5×10^5, the AAD increased from 3.3 % to 11.8 %. The typical value of $C = 10^5$ led the solution to not converge from about 2000 s onwards. For case 2, the best value was $C = 7.5 \times 10^4$ and changing it to 10^7 took the AAD from 4.5 % to 7.5 %. This indicated that both isothermal or mushy range materials are impacted by the porosity constant, although the first is more affected. Therefore, this parameter should always be checked when using the enthalpy-porosity scheme.
- Concerning the PCM properties: for case 1, altering from the average PCM density (912.5 kg/m^3) to the liquid PCM density (885 kg/m^3) cause a reduction of 0.5 % on the AAD. On the other hand, by changing the source of the coefficient of thermal expansion increased the AAD in 7.5 % (from $\beta = 0.000615\,\mathrm{K}^{-1}$ to $\beta = 0.00083\,\mathrm{K}^{-1}$). For case 2, the melting temperature also played a significant role, where the AAD increased in 6.9 % by changing from the liquidus temperature of case 1 (43.5 °C) to the average liquidus-solidus temperature of case 1 (45.85 °C). Hence, similarly to the porosity constant, the PCM properties should be validated with experimental results.
- Concerning the interaction between porosity constant and PCM properties: for the geometry of case 1 with the porosity constant (7.5×10^4) and PCM properties of case 2, a difference for the total melting time of 1400 s (26 %) was found. The maximum deviation was as large as 57.9 %, with an AAD for the liquid fraction of 38.0 %. In order to reduce this deviation to 5.7 %, the porosity constant should be adjusted to 7.5×10^6 in the new configuration. The opposite also revealed this issue, i.e. geometry of case 2 with the porosity constant and properties of case 1, with AAD of 23.4 %. Nevertheless, in this situation, it was not possible to assess the proper correction on the porosity constant, since the solution did not

converge for some C values that would tend to reduce the deviation. Therefore, the practice of validating the model with a PCM and using it with a distinct material or in a completely different geometry could lead to unclear results. In this sense, this praxis should be discouraged as possible, although experimental results may not be available for the desired condition.

- Concerning the relation between porosity constant and natural convection intensity: since the characteristic length is different for cases 1 and 2, which blocks a comparison of Ra, the natural convection intensity has been indirectly inferred through the assessment of velocity and convective coefficients. Notably, our findings suggest that, in instances of elevated natural convection intensities, a reduction in the porosity constant is recommended.

The results presented in this work may be of interest to those engaged with the simulation of LTES systems, when using the enthalpy-porosity scheme. In this case, quantitative outcomes should be used with caution whenever the validation condition significantly differs from that of utilization. However, further investigations with different materials and/or distinct geometries could be performed to endorse, or not, the findings of this work.

CRediT authorship contribution statement

Thiago Torres Martins Rocha: Conceptualization, Validation, Methodology, Writing – original draft. **Paulo Vinicius Trevizoli:** Formal analysis, Writing – review & editing, Supervision. **Raphael Nunes de Oliveira:** Writing – review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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References

[1] IRENA, Innovation Outlook: Thermal Energy Storage, 2020. https://www.irena.org/publications/2020/Nov/Innovation-outlook-Thermal-energy-storage.
[2] R. Senthil, Effect of charging of phase change material in vertical and horizontal rectangular enclosures in a concentrated solar receiver, Case Stud. Therm. Eng. 21 (2020), 100653, https://doi.org/10.1016/j.csite.2020.100653.
[3] H.M. Taqi Al-Najjar, J.M. Mahdi, Novel mathematical modeling, performance analysis, and design charts for the typical hybrid photovoltaic/phase-change material (PV/PCM) system, Appl. Energy 315 (2022), 119027, https://doi.org/10.1016/j.apenergy.2022.119027.
[4] X. Kong, L. Zhang, H. Li, Y. Wang, M. Fan, Experimental thermal and electrical performance analysis of a concentrating photovoltaic/thermal system integrated with phase change material (PV/T-CPCM), Sol. Energy Mater. Sol. Cells 234 (2022), 111415, https://doi.org/10.1016/j.solmat.2021.111415.
[5] P. Yan, W. Fan, Y. Han, H. Ding, C. Wen, A.F.A. Elbarghthi, Y. Yang, Leaf-vein bionic fin configurations for enhanced thermal energy storage performance of phase change materials in smart heating and cooling systems, Appl. Energy 346 (2023), 121352, https://doi.org/10.1016/j.apenergy.2023.121352.
[6] E. Inkeri, T. Tynjälä, M. Nikku, Numerical modeling of latent heat thermal energy storage integrated with heat pump for domestic hot water production, Appl. Therm. Eng. 214 (2022), 118819, https://doi.org/10.1016/j.applthermaleng.2022.118819.
[7] T.T.M. Rocha, M. Teggar, P.V. Trevizoli, R.N. de Oliveira, Potential of latent thermal energy storage for performance improvement in small-scale refrigeration units: A review, Renew. Sustain. Energy Rev. 187 (2023), 113746, https://doi.org/10.1016/j.rser.2023.113746.
[8] O. Arfi, E.H. Mezaache, A. Laouer, Numerical investigation of double layered PCM building envelope during charging cycle for energy saving, Int. Commun. Heat Mass Transf. 144 (2023), 106797, https://doi.org/10.1016/j.icheatmasstransfer.2023.106797.
[9] K.A.R. Ismail, F.A.M. Lino, P.L.O. Machado, M. Teggar, M. Arıcı, T.A. Alves, M.P. R. Teles, New potential applications of phase change materials: A review, J. Energy Storage. 53 (2022), 105202, https://doi.org/10.1016/j.est.2022.105202.
[10] S. Ahmed, A. Abderrahmane, A.M. Saeed, K. Guedri, A. Mourad, O. Younis, T. Botmart, N.A. Shah, Melting enhancement of PCM in a finned tube latent heat thermal energy storage, Sci. Rep. 12 (2022) 11521, https://doi.org/10.1038/s41598-022-15797-0.
[11] C. Zhang, Q. Sun, Y. Chen, Solidification behaviors and parametric optimization of finned shell-tube ice storage units, Int. J. Heat Mass Transf. 146 (2020), 118836, https://doi.org/10.1016/j.ijheatmasstransfer.2019.118836.
[12] F. He, R. Bo, C. Hu, X. Meng, W. Gao, Employing spiral fins to improve the thermal performance of phase-change materials in shell-tube latent heat storage units, Renew. Energy 203 (2023) 518–528, https://doi.org/10.1016/j.renene.2022.12.091.
[13] Z.J. Zheng, Y. Chen, Y. Sun, C. He, H. Yin, Y. Xu, Melting and solidification performance enhancement of phase change material in a square cavity by utilizing temperature-driven deformable fin, Appl. Therm. Eng. 226 (2023), 120313, https://doi.org/10.1016/j.applthermaleng.2023.120313.
[14] A. Tafone, R. Pili, M. Pihl Andersen, A. Romagnoli, Dynamic modelling of a compressed heat energy storage (CHEST) system integrated with a cascaded phase change materials thermal energy storage, Appl. Therm. Eng. 226 (2023), 120256, https://doi.org/10.1016/j.applthermaleng.2023.120256.
[15] A. Abderrahmane, M. Al-Khaleel, A. Mourad, H. Laidoudi, Z. Driss, O. Younis, K. Guedri, R. Marzouki, Natural convection within inversed T-shaped enclosure filled by nano-enhanced phase change material: numerical investigation, Nanomaterials 12 (2022) 2917, https://doi.org/10.3390/nano12172917.
[16] W. Al-Kouz, A. Aissa, S. Suriya Uma Devi, M. Prakash, L. Kolsi, H. Moria, W. Jamshed, O. Younis, Effect of a rotating cylinder on the 3D MHD mixed convection in a phase change material filled cubic enclosure, Sustain. Energy Technol. Assessm. 51 (2022), https://doi.org/10.1016/j.seta.2021.101879.
[17] T.T.M. Rocha, P.V. Trevizoli, R.N. de Oliveira, A timeline of the phase-change problem for latent thermal energy storage systems: a review of theoretical approaches from the 1970's to 2022, Sol. Energy. 250 (2023) 248–284. https://doi.org/10.1016/j.solener.2022.12.035.
[18] L. Chen, A. Fan, Effects of shell modifications and operational parameters on melting uniformity of a vertical multi-section shell-and-tube latent heat thermal energy storage unit, J. Energy Storage. 55 (2022), 105593, https://doi.org/10.1016/j.est.2022.105593.
[19] W. Ye, J.M. Khodadadi, Effects of arrow-shape fins on the melting performance of a horizontal shell-and-tube latent heat thermal energy storage unit, J. Energy Storage. 54 (2022), 105201, https://doi.org/10.1016/j.est.2022.105201.
[20] Z. Haddad, F. Iachachene, H.F. Oztop, Melting characteristics of organic phase change material in a wavy trapezoidal cavity, J. Mol. Liq. 332 (2021), 112132, https://doi.org/10.1016/j.molliq.2019.112132.
[21] H. Wang, H. Xu, C. Liu, Z. Zhang, X. Ling, F. Jiang, Numerical investigation of melting enhancement for paraffin in an innovative finned-plate latent heat thermal energy storage unit, J. Energy Storage. 43 (2021), 103222, https://doi.org/10.1016/j.est.2021.103222.
[22] L. Zhang, P. Yang, W. Li, J.J. Klemeš, M. Zeng, Q. Wang, A new structure of PCHE with embedded PCM for attenuating temperature fluctuations and its performance analysis, Energy 254 (2022) 124462.
[23] M. Ghalambaz, S.A.M. Mehryan, A. Veismoradi, M. Mahdavi, I. Zahmatkesh, Z. Kazemi, O. Younis, M. Ghalambaz, A.J. Chamkha, Melting process of the nano-enhanced phase change material (NePCM) in an optimized design of shell and tube thermal energy storage (TES): Taguchi optimization approach, Appl. Therm. Eng. 193 (2021), 116945, https://doi.org/10.1016/j.applthermaleng.2021.116945.
[24] A.M. Saeed, A. Abderrahmane, N.A.A. Qasem, A. Mourad, M. Alhazmi, S.E. Ahmed, K. Guedri, A numerical investigation of a heat transfer augmentation finned pear-shaped thermal energy storage system with nano-enhanced phase change materials, J. Energy Storage. 53 (2022), 105172, https://doi.org/10.1016/j.est.2022.105172.
[25] S.E. Ahmed, A. Abderrahmane, S. Alotaibi, O. Younis, R.A. Almasri, W.K. Hussam, Enhanced heat transfer for NePCM-melting-based thermal energy of finned heat pipe, Nanomaterials 12 (2021) 129, https://doi.org/10.3390/nano12010129.
[26] A. Mourad, A. Aissa, A.M. Abed, G.F. Smaisim, D. Toghraie, M.A. Fazilati, O. Younis, K. Guedri, A. Alizadeh, The numerical analysis of the melting process in a modified shell-and-tube phase change material heat storage system, J. Energy Storage. 55 (2022), 105827, https://doi.org/10.1016/j.est.2022.105827.
[27] Y. Ezzat, A.A. Abdel-Rehim, Numerical modelling of lauric acid phase change material using iterative and non-iterative time-advancement schemes, J. Energy Storage. 53 (2022), 105173, https://doi.org/10.1016/j.est.2022.105173.
[28] İ.G. Demirkıran, E. Cetkin, Computation time reduction of PCM melting process by changing modeling parameters, Numer. Heat Transf. Part A Appl. 83 (2022) 50–67, https://doi.org/10.1080/10407782.2022.2149229.
[29] V.R. Voller, C. Prakash, A fixed grid numerical modelling methodology for convection-diffusion mushy region phase-change problems, Int. J. Heat Mass Transf. 30 (1987) 1709–1719, https://doi.org/10.1016/0017-9310(87)90317-6.
[30] ANSYS FLUENT User's Guide, 15.0, ANSYS, Inc, 2013.

[31] A.C. Kheirabadi, D. Groulx, The effect of the mushy-zone constant on simulated phase change heat transfer, in: Int. Symp. Adv. Comput. Heat Transf., 2015: pp. 528–549. https://doi.org/10.1615/ICHMT.2015.IntSympAdvComputHeatTransf.460.
[32] H. Shmueli, G. Ziskind, R. Letan, Melting in a vertical cylindrical tube: Numerical investigation and comparison with experiments, Int. J. Heat Mass Transf. 53 (2010) 4082–4091, https://doi.org/10.1016/j.ijheatmasstransfer.2010.05.028.
[33] A. Marušić, D. Lončar, Experimental validation of high-temperature latent heat storage model using melting front propagation data, Appl. Therm. Eng. 164 (2020) 114520.
[34] M. Fadl, P.C. Eames, Numerical investigation of the influence of mushy zone parameter Amush on heat transfer characteristics in vertically and horizontally oriented thermal energy storage systems, Appl. Therm. Eng. 151 (2019) 90–99, https://doi.org/10.1016/j.applthermaleng.2019.01.102.
[35] R. De Césaro Oliveski, F. Becker, L.A.O. Rocha, C. Biserni, G.E.S. Eberhardt, Design of fin structures for phase change material (PCM) melting process in rectangular cavities, J. Energy Storage. 35 (2021) 102337.
[36] C.J. Troxler, T.B. Freeman, R.M. Rodriguez, S.K.S. Boetcher, Experimental and numerical investigation of lauric acid melting at suboptimal inclines, ASME Open J. Eng. 2 (2023), https://doi.org/10.1115/1.4056348.
[37] L.-L. Tian, X. Liu, S. Chen, Z.-G. Shen, Effect of fin material on PCM melting in a rectangular enclosure, Appl. Therm. Eng. 167 (2020) 114764.
[38] C. Reichl, S. Both, P. Mascherbauer, J. Emhofer, Comparison of two CFD approaches using constant and temperature dependent heat capacities during the phase transition in PCMs with experimental and analytical results, Processes. 10 (2) (2022) 302.
[39] J. Vogel, A. Thess, Validation of a numerical model with a benchmark experiment for melting governed by natural convection in latent thermal energy storage, Appl. Therm. Eng. 148 (2019) 147–159, https://doi.org/10.1016/j.applthermaleng.2018.11.032.
[40] L. Yang, X. Cao, N. Zhang, B. Xiang, Z. Zhang, B. Qian, Thermal reliability of typical fatty acids as phase change materials based on 10,000 accelerated thermal cycles, Sustain. Cities Soc. 46 (2019), 101380, https://doi.org/10.1016/j.scs.2018.12.008.
[41] M.S. Mahdi, H.B. Mahood, A.A. Khadom, A.N. Campbell, Numerical simulations and experimental verification of the thermal performance of phase change materials in a tube-bundle latent heat thermal energy storage system, Appl. Therm. Eng. 194 (2021), 117079, https://doi.org/10.1016/j.applthermaleng.2021.117079.
[42] Y. Yuan, X. Cao, B. Xiang, Y. Du, Effect of installation angle of fins on melting characteristics of annular unit for latent heat thermal energy storage, Sol. Energy 136 (2016) 365–378, https://doi.org/10.1016/j.solener.2016.07.014.
[43] X. Cao, Y. Yuan, B. Xiang, F. Highlight, Effect of natural convection on melting performance of eccentric horizontal shell and tube latent heat storage unit, Sustain. Cities Soc. 38 (2018) 571–581, https://doi.org/10.1016/j.scs.2018.01.025.
[44] L. Kalapala, J.K. Devanuri, Energy and exergy analyses of latent heat storage unit positioned at different orientations – an experimental study, Energy 194 (2020), 116924, https://doi.org/10.1016/j.energy.2020.116924.
[45] H. Shokouhmand, B. Kamkari, Experimental investigation on melting heat transfer characteristics of lauric acid in a rectangular thermal storage unit, Exp. Therm Fluid Sci. 50 (2013) 201–212, https://doi.org/10.1016/j.expthermflusci.2013.06.010.
[46] C. Liu, D. Groulx, Experimental study of the phase change heat transfer inside a horizontal cylindrical latent heat energy storage system, Int. J. Therm. Sci. 82 (2014) 100–110, https://doi.org/10.1016/j.ijthermalsci.2014.03.014.
[47] M.Y. Shaker, A.A. Sultan, E.A. El Negiry, A. Radwan, Melting and solidification characteristics of cylindrical encapsulated phase change materials, J. Energy Storage. 43 (2021), 103104, https://doi.org/10.1016/j.est.2021.103104.
[48] H. Li, N. Wang, S. He, M. Gao, Effect of inner-tube spacing on charging and discharging performance of latent energy storage heat exchangers, Appl. Therm. Eng. 216 (2022), 119112, https://doi.org/10.1016/j.applthermaleng.2022.119112.
[49] M. Zeneli, A. Nikolopoulos, S. Karellas, N. Nikolopoulos, Numerical methods for solid-liquid phase-change problems, in: Ultra-High Temp. Therm. Energy Storage, Transf. Convers., LTD, 2021: pp. 165–199. https://doi.org/10.1016/B978-0-12-819955-8.00007-7.
[50] B. Kamkari, H. Shokouhmand, F. Bruno, Experimental investigation of the effect of inclination angle on convection-driven melting of phase change material in a rectangular enclosure, Int. J. Heat Mass Transf. 72 (2014) 186–200, https://doi.org/10.1016/j.ijheatmasstransfer.2014.01.014.
[51] I.B. Celik, U. Ghia, P.J. Roache, C.J. Freitas, H. Coleman, P.E. Raad, Procedure for estimation and reporting of uncertainty due to discretization in CFD applications, J. Fluids Eng. Trans. ASME. 130 (2008) 0780011–0780014, https://doi.org/10.1115/1.2960953.
[52] P. Jany, A. Bejan, Scaling theory of melting with natural convection in an enclosure, Int. J. Heat Mass Transf. 31 (1988) 1221–1235, https://doi.org/10.1016/0017-9310(88)90065-8.
[53] M. Farid, Y. Kim, T. Honda, A. Kanzawa, The role of natural convection during melting and solidification of PCM in a vertical cylinder, Chem. Eng. Commun. 84 (1989) 43–60, https://doi.org/10.1080/00986448908940334.
[54] M. Zeneli, I. Malgarinos, A. Nikolopoulos, N. Nikolopoulos, P. Grammelis, S. Karellas, E. Kakaras, Numerical simulation of a silicon-based latent heat thermal energy storage system operating at ultra-high temperatures, Appl. Energy 242 (2019) 837–853, https://doi.org/10.1016/j.apenergy.2019.03.147.